NITISH KUMAR

Software Engineer | Python Backend | AI/ML Systems & Agentic AI

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PROFESSIONAL SUMMARY

Software engineer with approximately 3.5 years of experience building Python backend services, REST APIs, LLM-integrated applications, and agentic workflows. Hands-on experience with Django, Flask, FastAPI, RAG, tool-using agents, browser automation, and deployment automation. Experienced in client-facing delivery, technical requirement gathering, and coordinating Python development work. Interested in building reliable AI systems and production-grade software infrastructure.

TECHNICAL SKILLS

Programming: Python, JavaScript

Backend & APIs: Django, Django REST Framework, Flask, FastAPI, Node.js, REST APIs, microservices

AI & LLM Systems: LLM API integration, RAG, embeddings, PGVector, Hugging Face Transformers, LangGraph, CrewAI, MCP, Playwright

ML & Data: TensorFlow, PyTorch, scikit-learn, Transformers, OpenCV, Pandas, NumPy

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis

Cloud & Delivery: AWS EC2/S3, Docker, GitHub Actions, Git, CI/CD, Linux

Engineering Practices: API performance optimization, logging, observability, output validation, retries, Agile, client collaboration

PROFESSIONAL EXPERIENCE

Senior Python Developer — Snakescript Solutions LLP

Apr 2023 – Sep 2026

- Designed, developed, and deployed API-first backend services using Django, Flask, and FastAPI for AI-enabled products and client applications.
- Built backend logic and API orchestration for AI voice-agent workflows and real-time call-handling experiences.
- Led client discussions and translated requirements into deliverable technical solutions; reported a 95% client satisfaction rate.
- Coordinated Python development work using Agile practices, contributing to a reported 30% reduction in delivery turnaround time.
- Reduced LLM API costs by a reported 35% through prompt optimization, caching, and response validation.
- Improved API modularity and performance through microservice design and optimization, reporting a 40% reduction in API latency.
- Automated deployment and AI workflows using GitHub Actions, contributing to a reported 50% reduction in deployment cycles.

SELECTED PROJECTS

Autonomous Browser Automation Agent

Python, Playwright, MCP, Docker, noVNC, LLM APIs

- Built an AI-powered agent that navigates web applications and performs multi-step tasks from natural-language instructions.
- Integrated MCP tools for structured communication between the model and browser environment; added session memory and retry handling.
- Implemented live browser monitoring and manual takeover through noVNC, and asynchronous Playwright operations for workflow execution.

AI-Powered Conversational Agent

Python, Django, OpenAI API, RAG, Redis, PGVector

- Developed a conversational application with context handling and conversation history for domain-specific questions.
- Implemented retrieval-augmented responses and integrated speech-to-text/text-to-speech capabilities; reported a 35% increase in successful query resolutions.

AI Website Generation Platform (Lovable-style clone)

Python, CrewAI, RAG, PGVector, LangSmith

- Built an agentic workflow for prompt analysis, UI planning, component generation, and final HTML generation.
- Used PGVector retrieval for reusable UI components and design patterns, and LangSmith for tracing, debugging, and evaluation.
- Added output validation and refinement steps to improve the consistency of responsive, browser-ready HTML.

SELECTED PROJECTS — CONTINUED

E-commerce Platform with AI-Powered Recommendations

Django, PostgreSQL, Stripe API, TensorFlow Recommenders

- Developed e-commerce functionality covering authentication, product management, and payment integration.
- Implemented collaborative-filtering recommendations; reported a 20% increase in average order value.
- Optimized database queries and caching, reporting a 40% reduction in page-load time.

Personal AI Voice Assistant

Python, Django, Vapi, Make.com, PythonAnywhere

- Built a real-time voice assistant with call handoff and automated call-summary emails.
- Connected the voice workflow to a Django backend and automation integrations.

ADDITIONAL ENGINEERING EXPERIENCE

- API development and integration across third-party services and AI-enabled workflows.
- Deployment automation and production troubleshooting in Linux-based environments.
- Client and stakeholder communication, requirements gathering, and technical delivery coordination.

EDUCATION

Master of Computer Applications (MCA), Artificial Intelligence & Machine Learning

Chandigarh University | 2021–2023 | CGPA: 7.7/10

ADDITIONAL INFORMATION

Professional strengths: Analytical problem-solving, client communication, cross-functional collaboration, team coordination, and continuous learning.

Interests: Chess, badminton, reading, and public speaking.